



Is the door really open? A contingent model of boundary spanning behavior and abusive supervisory behavior

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ABSTRACT

Although research suggests that subordinates' boundary spanning may result in unintended negative responses from their supervisors, little is known about which supervisors are likely to react adversely and which are not. We draw on self-protection theory to answer this question. Specifically, we argue that when supervisors experience low levels of managerial self-efficacy, boundary spanning behavior threatens their perceived work status within the team, resulting in abusive supervisory behavior. The results of two field survey studies and an experimental study support our hypotheses. Our research identifies managerial barriers that may hinder boundary spanning behavior at work and provides invaluable insights into how upper-level organizations may create constructive work contexts that encourage employees to engage in such behavior.

1. Introduction

The growing challenges of task complexity and environmental volatility have made it essential for teams to establish and maintain connections with key external parties to obtain resources and support, which is referred to as boundary spanning behavior (Ancona & Caldwell, 1992; Marrone et al., 2007). Studies have shown that team members' boundary spanning behavior contributes to team effectiveness (e.g., Faraj & Yan, 2009; Hargadon, 2002; Marrone et al., 2007; Somech & Khalaili, 2014; Xie et al., 2022). Although there has been extensive scholarly work supporting boundary spanning, particularly supervisors' supportive and coaching behaviors (e.g., Ancona & Bresman, 2007; Marrone et al., 2022), recent studies have suggested that team members' boundary spanning might result in adverse responses from supervisors, such as receiving less favorable evaluations (Clement et al., 2018) or being undermined by their supervisors (Mell et al., 2022).

These findings indicate that supervisors may become internal barriers preventing team members from engaging in boundary spanning behavior. However, do all supervisors view subordinates' boundary spanning unfavorably? Although some research has explored how subordinates' upward advice-seeking behavior may influence supervisors' interpretation of and negative reactions toward their boundary spanning behavior (Mell et al., 2022), few studies have considered supervisors' characteristics as a critical determining factor in how supervisors

appraise and react to subordinates' boundary spanning. This omission is consequential. The leadership research has suggested that the extent to which supervisors, as the "gatekeepers" of their teams, endorse or reject team members' boundary spanning has substantial impacts on their subsequent engagement in such behavior (e.g., Lord et al., 2017). Therefore, it is essential to identify which supervisors tend to react negatively to such behavior, as doing so will enable the development of more effective coping strategies for eliminating the hurdles to subordinates' engagement in boundary spanning.

In light of these assertions, this study attempts to unravel when and why supervisors react negatively to their subordinates' boundary spanning behavior. Specifically, we draw on self-protection theory (Sedikides, 2012; Sedikides & Alicke, 2012) to explore our central question. Self-protection theory posits that individuals are inclined to perceive threats in stimuli that could potentially harm their positive self-concepts, and status is one of the people's primary concerns (Alicke & Sedikides, 2009; Sedikides, 2012). Studies have indicated that managerial positions with superior hierarchical power are the primary source of managers' self-concept (e.g., Fast et al., 2014; Thoits, 1991). To preserve a positive self-image (i.e., high status), supervisors are expected to demonstrate substantial competence in fulfilling essential leadership functions such as boundary spanning (Fast & Chen, 2009; Hollander, 1958; Morgeson et al., 2010). Accordingly, subordinates who frequently engage in boundary spanning may prevent supervisors from

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representing their team and cause them to lose advantageous positions obtained by liaising with critical external players, and thus these actions may be perceived as a status threat (Fast et al., 2014; Hollander, 1958).

However, not all supervisors perceive subordinates' boundary spanning behavior as a threat to their status. Self-protection theory suggests that the extent to which individuals feel threatened by stimuli that undermine their core self-interests, such as status, hinges on their self-evaluation (Sedikides, 2012; Sedikides & Alicke, 2012; Tice, 1991). Those with a lower self-evaluation are more likely to feel threatened. For supervisors, self-view is closely related to how well they perform their managerial roles (Fast et al., 2014; Hollander, 1958). We argue that managerial self-efficacy, which reflects managers' confidence in their capabilities to fulfill the demands and expectations associated with managerial roles (Bandura, 1982; Fast et al., 2014; Judge & Bono, 2001), is an essential manifestation of supervisors' self-evaluation in the work context (Chen et al., 2004; Hannah et al., 2008). Hence, supervisors with lower managerial self-efficacy are more likely than those with higher self-efficacy to perceive a status threat when confronted with subordinates who frequently cross team boundaries.

Moreover, according to self-protection theory, the perceptions of potential or actual threats can trigger rapid self-protection strategies (Roese & Olson, 2007; Sedikides, 2012; Sedikides & Alicke, 2012). Given that supervisors have formal authority over subordinates, abusive supervisory behavior, defined as supervisors' interpersonal mistreatment of subordinates (Tepper, 2000), can serve as an effective strategy for eliminating a potential threat to the status imposed by subordinates. Such behavior could increase subordinates' sensitivity to supervisors' control and influence (Ferris et al., 2007; Tepper et al., 2012), preventing the targeted followers from engaging in frequent boundary spanning in the future. Therefore, we propose that supervisors may engage in abusive supervisory behavior when they perceive their subordinates' boundary spanning as a threat to their status. In addition, as stronger personal efficacy usually leads to greater confidence in handling stressful situations (e.g., Bandura, 2010; Gecas, 1989), we posit that managerial self-efficacy attenuates the positive association between the supervisor's perceived status threat and abusive responses. Fig. 1 illustrates our theoretical model. We conduct two survey studies and one experimental study with complementary methodological approaches to provide a robust test of our hypotheses.

Our research makes several contributions. First, we extend the literature on boundary spanning by identifying which supervisors tend to react negatively to subordinates' boundary spanning behavior. Supervisors can be the primary barrier to crossing team boundaries. Research has indicated that supervisors generally perceive subordinate boundary spanning as leading to territorial loss, and thus it promotes negative supervisory reactions (e.g., Mell et al., 2022). However, less is known about which supervisors are likely to have adverse psychological and behavioral responses to subordinate boundary spanning. In this study, we recognize that supervisors with low managerial self-efficacy are more likely to perceive threats than those with high managerial self-efficacy and thus are more likely to demonstrate abusive supervisory behavior in response to subordinates' boundary spanning.

Second, we advance the literature by revealing a critical underlying mechanism linking subordinates' boundary spanning and negative

supervisory responses. Studies have shown that supervisors' concerns about losing group territorial control prompt them to undermine members who frequently cross boundaries (Mell et al., 2022). In this study, we treat boundary spanning as a representative leadership function (Morgeson et al., 2010), and the extent to which supervisors fulfill the demand of managing the boundary is closely associated with their status or influence at work (Fast et al., 2014; Hollander, 1958). Our study reveals that perceived status threat is an essential underlying mechanism for explaining supervisors' negative reactions to subordinates' boundary spanning. Third, we expand self-protection theory by utilizing this fundamental framework to delineate the complexity of power holders' (i.e., supervisors') self-protection process when they encounter negative stimuli caused by their direct subordinates. We identify status threat as an essential type of self-threat and abusive behavior as an aggressive self-protection strategy against potential self-threat, particularly for supervisors.

2. Theory and hypotheses

2.1. Subordinate boundary spanning behavior and supervisors' perceptions of status threat

Boundary spanning behavior refers to actions by team members that build connections and manage interactions with external players with the aim of obtaining critical resources and support for the team (Ancona & Caldwell, 1992; Faraj & Yan, 2009; Marrone et al., 2007). Ancona (1990) identified various boundary spanning activities in her seminal article, such as representing the team for external parties, coordinating work activities with members from other groups, or seeking information from external experts. Boundary spanning behavior, which aids teams in promptly acquiring critical task-related information and valuable support (Ancona et al., 2002; Ancona & Bresman, 2007), has been widely recognized as increasing team effectiveness and improving performance (e.g., Bresman, 2010; Faraj & Yan, 2009; Joshi et al., 2009; Marrone et al., 2007; Xie et al., 2022).

However, supervisors may not regard such behavior as positive (e.g., Clement et al., 2018; Mell et al., 2022). In this study, we rely on self-protection theory to explain *when* and *why* supervisors react negatively to subordinates' boundary spanning behavior. The key insight of self-protection theory is that people try to maintain their positive self-view (Sedikides, 2012). When elements of their core self-concept, such as their social status, are challenged, they take action to eliminate the perceived threat (Alicke & Sedikides, 2009; Sedikides, 2012). As formal leaders of work teams, supervisors' self-concept is typically derived from managerial roles with high hierarchical authority (Fast et al., 2014; Thoits, 1991). Supervisors desire to maintain their position within the team and the prominence of their contributions, and may fear being considered incompetent and losing their influence (Cuddy et al., 2011; Fast et al., 2014). Thus, they will be alert to any actions of their subordinates that could potentially harm their superior status (Pratto et al., 2006; Sidanius & Pratto, 2001). We argue that boundary spanning by subordinates can be a behavioral cue that challenges and thus threatens their supervisors' current status.

According to functional leadership research, boundary spanning is a representative team function for which team leaders are held responsible (e.g., Hogg et al., 2012; Morgeson et al., 2010). Team members' boundary spanning behavior undermines supervisors' advantageous position as the link between the team and the external environment (Burt, 2004; Marrone, 2010). This may lead to supervisors experiencing barriers when soliciting critical task-related resources and may make their contributions to the team less prominent. Moreover, supervisors are less able to speak on behalf of their teams when their direct subordinates frequently cross team boundaries. Consequently, supervisors will lose crucial connections with key external players such as clients, experts, and power holders from other coordinated teams or units (Ancona & Caldwell, 1992; Marrone, 2010). In short, frequent boundary

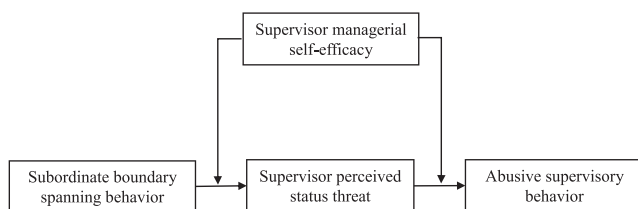


Fig. 1. Theoretical model of subordinate boundary spanning behavior and abusive supervisory behavior.

spanning by subordinates may cause supervisors to appear incompetent and unable to carry out their managerial role expectations, leading to diminished status within the group (Fast et al., 2014; Hollander, 1958) and perceptions of status threat.

2.2. Contingent effect of managerial self-efficacy

Self-protection theory implies that not all supervisors will perceive a status threat when confronted with subordinates' boundary spanning behavior. Rather, their sensitivity and subsequent responses to such behavior vary. According to this theory, how people interpret and react to stimuli largely depends on their level of self-evaluation (Hepper et al., 2010; Sedikides, 2012; Tice, 1991). Those with low self-evaluation will be more sensitive to potential negative outcomes than those with high self-evaluation, who will in turn be more interested in the potential positive outcomes (Ferris et al., 2011; Hepper et al., 2010; Tice, 1991).

In the workplace, supervisors' self-view is primarily associated with their enacted managerial positions (Fast et al., 2014; Thoits, 1991). We propose that their confidence in their performance of leadership functions or managerial role expectations, referred to as managerial self-efficacy, reflects their current level of self-evaluation (Bandura, 1982; Chen et al., 2004; Fast et al., 2014). In addition, supervisors' managerial self-efficacy also reflects their confidence in maintaining an advantageous status within the team and thus should influence their interpretations of subordinates' boundary spanning behavior.

We therefore introduce supervisor managerial self-efficacy as a contingency factor in the relationship between subordinate boundary spanning behavior and supervisor perceived status threat. Supervisors with low levels of managerial self-efficacy will feel anxious about being perceived as less competent and influential within their teams (Fiske, 2018; Fast et al., 2014). Self-protection theory suggests that they will be particularly concerned about protecting themselves from devaluation and thus more likely to feel threatened by stimuli (such as subordinate boundary spanning behavior) that challenge the status hierarchy.

However, those with high levels of managerial self-efficacy may believe that they can effectively deal with any eventualities and will be confident about demonstrating their efficacy within the team (Fast et al., 2014). They may prefer to pursue opportunities that strengthen their existing positions within the hierarchy (Ferris et al., 2011; Hepper et al., 2010; Tice, 1991) rather than focusing on actions that may threaten their positions, such as subordinate boundary spanning behavior. Furthermore, supervisors with high levels of managerial self-efficacy are more approach-oriented and may even view subordinate boundary spanning behavior favorably because it can improve team performance, thus fulfilling their task goals. Drawing on this rationale, we propose the following hypothesis.

Hypothesis 1. *Supervisor managerial self-efficacy moderates the relationship between subordinate boundary spanning behavior and supervisor perceived status threat, such that the relationship is only positive when supervisor managerial self-efficacy is low (vs. high).*

2.3. Supervisor perceived status threat and abusive supervisory behavior

Self-protection theory can also help to explain why supervisors may react negatively to a subordinate whom they perceive as threatening their status. This theory suggests that individuals will actively use self-protection strategies if their positive self-regard is threatened (Sedikides, 2012; Sedikides & Alicke, 2012). Fast and Chen (2009) demonstrated that individuals with superior authority at work (i.e., supervisors) are more prone to engage in aggressive behaviors when they feel incapable of meeting the expectations of their high-power roles and have self-threat perceptions. Moreover, research has also indicated that some destructive leadership behaviors, such as abusive supervisory behavior, can be used strategically to influence subordinates and achieve the supervisors' personal objectives (e.g., Ferris et al., 2007; Walter

et al., 2015). Therefore, we argue that abusive supervisory behavior can be an effective self-protection strategy that reduces supervisor perceived status threat for the following reasons.

First, abusive supervisory behavior, such as pointing out subordinates' faults or failures and making unfavorable comments about them in public, can be used deliberately to diminish subordinates' authority and influence, making them more susceptible to the supervisors' influence and control (Ferris et al., 2007; Hu & Liu, 2017). As a result, supervisors may prevent subordinates from frequently spanning boundaries to protect themselves from potential status threats.

Second, abusive supervisory behavior also can be used strategically to manage others' impressions and thus fortify supervisors' image of authority and status (e.g., Khan et al., 2018; Tedeschi & Felson, 1994). Hu and Liu (2017) noted that status concerns can be a significant psychological driver of abusive behavior. They found that supervisors engage in abusive supervisory behavior when motivated to enhance their status. Therefore, we argue that to retain a high status at work, supervisors who perceive a status threat from a direct subordinate are likely to engage in abusive supervisory behavior toward the subordinate. Based on this rationale, we assume that supervisors utilize abusive behavior as an effective self-protection strategy when they feel that their status is threatened. From these theoretical arguments, we propose **Hypothesis 2**.

Hypothesis 2. *Supervisors' perceptions of status threat are positively related to abusive supervisory behavior.*

In addition, we contend that supervisors' managerial self-efficacy may further attenuate the positive association between supervisor perceived status threat and abusive supervision. Although supervisors are more prone to engage in abusive responses when they perceive status threats, those with greater managerial self-efficacy may be less inclined to do so. Research has indicated that having high personal efficacy generally leads to a stronger belief in one's ability to control the attainment of goals and greater confidence in handling stressful situations (e.g., Bandura, 2010; Gecas, 1989). In this case, supervisors with high managerial self-efficacy will have greater self-confidence and a greater sense of personal security when confronted with status-threatening situations, making them less likely to react in an abusive way. In contrast, supervisors with low managerial self-efficacy tend to feel insecure when they encounter situations that threaten their status. Therefore, they are more prone to wield formal authority and engage in abusive reactions to eliminate the perceived status threat. Following these theoretical rationales, we propose **Hypothesis 3**.

Hypothesis 3. *Supervisor managerial self-efficacy moderates the positive relationship between supervisor perceived status threat and abusive supervisory behavior, such that the relationship is only positive when supervisor managerial self-efficacy is low (vs. high).*

Combining these arguments and rationale, we put forth the following moderated mediation hypothesis.

Hypothesis 4. *Supervisor managerial self-efficacy moderates the indirect effect of subordinate boundary spanning behavior on abusive supervisory behavior via supervisor perceived status threat, such that the indirect effect is only positive when supervisor managerial self-efficacy is low (vs. high).*

3. Overview of studies

We conducted three studies with complementary methodological approaches to test our proposed effects. Study 1 was a two-wave, multi-source survey involving participants from various industries and organizations and provided initial support for our proposed relationships. In Study 2, we constructively replicated the findings of Study 1 by conducting a two-wave, multi-source field study in two different organizations and, more importantly, ruled out confounding influences and alternative explanations. Finally, we conducted a vignette-style

experiment (i.e., Study 3) with full-time employees to obtain stronger causal inferences for our hypothesized relationships while preserving a relatively high degree of realism (Aguinis & Bradley, 2014). The different designs of these studies enabled us to conduct a rigorous and thorough examination of the focal relationships and thus validate our hypotheses both internally and externally.

3.1. Study 1: Methods

3.1.1. Participants and procedure

The participants in Study 1 were full-time managerial employees enrolled in a part-time MBA program at a large university in eastern China. They were invited to voluntarily participate in our study in exchange for course credit and were asked to provide a list of at least three randomly selected team members under their direct supervision as additional participants. Before they delivered the list of names, to ensure that the leaders and the listed members worked as a team, we asked the team leaders to briefly describe their teams' function and team members' daily work, and we clearly stated that participants in this study should come from the team that they were currently leading. Our sample initially consisted of 114 supervisors and 472 employees under their immediate supervision. Before collecting the data, we assured them of the voluntary and confidential nature of their participation.

We collected data from the sources at two time points to reduce common method bias (Podsakoff et al., 2012). At the beginning of the course, we organized a briefing session to explain the general research purposes and procedures without disclosing any specific predictions. The participants then completed an online survey that collected their demographic information. In addition, the supervisors provided data on the types of companies and industries they operated in. About 2 weeks later, at Time 1, we separately e-mailed online survey links to supervisors and subordinates. The subordinates completed the boundary spanning behavior measurement instrument, and the supervisors completed measures of managerial self-efficacy and perceived status threat for each participating subordinate. After another 2 weeks, at Time 2, the subordinates completed another online survey that assessed abusive supervisory behavior. To show our appreciation, we provided all of the participants with souvenir pens displaying their signatures.

After removing surveys with incomplete responses across the two waves, we obtained usable data from 99 supervisors and 320 immediate subordinates across various industries, including information technology, manufacturing, finance, services, real estate, and others, yielding an 86.8% response rate for supervisors and 67.8% for subordinates. Of the 99 supervisors, 69.7% were male, and 96.0% had a college education or above; they had an average age of 34.7 years ($SD = 4.0$) and an average organizational tenure of 7.1 years ($SD = 4.8$). Of the 320 subordinates, 55.0% were male, and 86.9% had a college or higher education; they had an average age of 29.3 years ($SD = 4.6$) and an average organizational tenure of 3.6 years ($SD = 3.1$). The average relationship tenure between the supervisors and subordinates was 31.9 months ($SD = 32.0$).

3.1.2. Measures

All of the measures used in Study 1 were translated from English into Chinese following Brislin's (1980) translation-back translation procedure.

Subordinate boundary spanning behavior. We measured subordinate boundary spanning behavior using the 15-item scale from Ancona and Caldwell (1992). The subordinates reported the frequency with which they engaged in these behaviors on a 7-point scale (1 = *never* and 7 = *always*, $\alpha = 0.97$). Sample items include "Persuade others to support the team's decisions," "Keep other groups in the company informed of your team's activities," and "Procure things that the team needs from other groups or individuals in the company."

Supervisor managerial self-efficacy. We measured supervisor managerial self-efficacy using an 8-item scale from Fast et al. (2014). To

ensure that the measure assessed supervisors' sense of competence in the management position, we also specifically asked the supervisors to answer the items as they related to their own managerial domain in the team that they led, which is in line with the statement from Fast et al. (2014). The participants rated their agreement with each statement on a 7-point response scale (1 = *strongly disagree* and 7 = *strongly agree*, $\alpha = 0.94$). Sample items include "I believe I can succeed at almost any endeavor to which I set my mind," "When facing difficult tasks, I am certain that I will accomplish them," and "I am confident that I can perform effectively on many different tasks."

Supervisor perceived status threat. We measured supervisor perceived status threat with a 3-item scale adapted from Burris (2012), which was originally used to measure supervisors' feelings of threat. In Burris's (2012) paper, supervisors were asked to report to what extent their competence or ability to perform would be questioned and to what extent they would lose status in the eyes of others when encountering subordinates' voice behavior. Studies have demonstrated that individuals' competence or ability to perform serves as the primary basis of their intra-group status (Magee & Galinsky, 2008). Thus, we believe that Burris's (2012) measure fits well with the supervisor perceived status threat examined in this study. We dropped one item (i.e., "Others will judge my plan to be faulty if my superiors hear this person's comments") that was deemed non-applicable to assessing supervisors' reactions to subordinates' boundary spanning behavior. The participants reported the extent to which each statement described them via a 7-point response scale (1 = *strongly disagree* and 7 = *strongly agree*, $\alpha = 0.86$). The items are "I will lose status in the team because of this subordinate," "My ability to lead the team will be questioned because of this subordinate," and "Others will doubt my competence because of this subordinate."

Abusive supervisory behavior. We measured abusive supervisory behavior using the 15-item scale from Tepper (2000). The subordinates indicated the frequency with which their supervisors displayed each type of abusive behavior toward them during workdays on a 7-point scale (1 = *never* and 7 = *always*, $\alpha = 0.96$). Sample items are "Ridicules me" and "Puts me down in front of others."

Control variables. First, research has suggested that organizational norms regarding hostile or aggressive behaviors may have an impact on abusive supervisory behavior (e.g., Tepper et al., 2017; Zhang & Bednall, 2016). To account for the influence of organizational norms, we controlled for company type.¹ The literature indicates that organizations with higher power distance values are more tolerant of unequal power differences and have less dissent (Javidan et al., 2006), and this might promote the acceptance of destructive leadership behaviors (Thoroughgood et al., 2018). The cultural value of power distance usually varies across the three types of firms in China due to the differences in their ownership structures (e.g., Shaheer et al., 2019; Wang, 2002).

Second, because the acceptance of supervisors' abusive behaviors may vary across different industries (e.g., Tepper, 2007; Thoroughgood et al., 2018), we also controlled for industry.² Further, the results of a meta-analysis revealed that subordinates' perceptions of abusive supervision were weakly associated with both subordinates' and supervisors' demographic profiles (e.g., age, gender, education, tenure), as well as their relationship tenure (Mackey et al., 2017). Our analyses also

¹ We created two dummy variables for three company types: private company (70.7%, *company dummy 1* = 1, *company dummy 2* = 0); state-owned company (17.2%, *company dummy 1* = 0, *company dummy 2* = 0); and foreign-invested company (12.1%, *company dummy 1* = 0, *company dummy 2* = 1).

² We created three dummy variables for the four main industries in our sample: information technology (25.3%, *industry dummy 1* = 0, *industry dummy 2* = 1, *industry dummy 3* = 0), manufacturing (23.2%, *industry dummy 1* = 1, *industry dummy 2* = 0, *industry dummy 3* = 0), finance (20.2%, *industry dummy 1* = 0, *industry dummy 2* = 0, *industry dummy 3* = 1), and others (31.3%, *industry dummy 1* = 0, *industry dummy 2* = 0, *industry dummy 3* = 0).

showed that none of these demographic control variables significantly altered our results. Therefore, based on recommendations to exclude controls when they do not change the results of the analysis (Becker, 2005; Carlson & Wu, 2012), we did not retain any demographic controls in our final analysis. The analyses with and without control variables produced virtually identical results.

3.1.3. Analytical strategy

Due to the multilevel nature of the data (i.e., subordinates were nested with supervisors), we conducted a two-level path analysis within the random slope framework of multilevel structural equation modeling (MSEM) (Liu et al., 2012; Preacher et al., 2016) using Mplus 8.3 (Muthén & Muthén, 1998–2017). At the within-dyad level, supervisor perceived status threat was modeled as a function of subordinate boundary spanning, and abusive supervisory behavior was modeled as a function of supervisor perceived status threat and subordinate boundary spanning. At the between-dyad level, we included supervisor managerial self-efficacy as a predictor of the within-dyad random slope of subordinate boundary spanning behavior and supervisor perceived status threat. We also included supervisor managerial self-efficacy as a predictor of the within-dyad random slope of subordinate boundary spanning and abusive supervisory behavior in addition to the within-dyad random slope of supervisor perceived status threat and abusive supervisory behavior. We group-mean centered the within-dyad level variables and grand-mean centered the supervisor level variable (i.e., managerial self-efficacy) (Hofmann & Gavin, 1998). We also included the group-mean of subordinate boundary spanning at the between level in the estimation of the random intercept of the endogenous variable in our model (Hofmann & Gavin, 1998). We used Snijders and Bosker's (1994) formulas to calculate *pseud-R*² for predicting the outcome variables in our MSEM. We tested the proposed conditional indirect effects using a Monte Carlo simulation procedure on the R software platform (Preacher & Selig, 2012).

3.1.4. Study 1: Results

3.1.4.1. Descriptive statistics and confirmatory factor analyses. We first estimated null models with only supervisor perceived status threat or abusive supervisory behavior as the dependent variable. The results showed significant between-team variances in supervisor perceived status threat ($\chi^2 = 532$, $df = 98$, $p < .001$; ICC1 = 0.58, indicating that 58% of variance resided between teams) and abusive supervisory behavior ($\chi^2 = 256$, $df = 98$, $p < .001$; ICC1 = 0.33, indicating that 33% of variance resided between teams). The descriptive statistics and correlations of the study variables are presented in Table 1.

We examined the distinctiveness of our main constructs by conducting multilevel confirmatory factor analyses (CFA) using the procedures recommended by Dyer and colleagues (2005). We used Mplus 8.3 (Muthén & Muthén, 1998–2017) to test the model in which supervisor managerial self-efficacy was treated as a group-level factor, and

subordinate boundary spanning behavior, supervisor perceived status threat, and abusive supervisory behavior was treated as both within-dyad and group-level factors. Previous literature suggested that it is not uncommon to encounter the problem of model non-convergence when conducting multilevel CFAs (Breidenthal et al., 2020; Marsh et al., 2009). Thus, to achieve a more optimal variable-to-sample-size ratio and minimize parameter estimate instability, we used the item-to-construct-balance parceling method (Bagozzi & Edwards, 1998; Little et al., 2002). We assigned subordinate boundary spanning and supervisor managerial self-efficacy items into four parcels, respectively and abusive supervisory behavior items into five parcels by sequentially averaging the aggregated items with the highest and lowest loadings (Little et al., 2002; Little et al., 2013). Results showed that the hypothetical four-factor baseline model fit the data well ($\chi^2 = 166.83$, $df = 149$, $p = .15$; CFI = 1.00, RMSEA = 0.02, SRMR_(within) = 0.03, SRMR_(between) = 0.07). This model fitted the data better than alternative models when the following variables were combined: (1) subordinate boundary spanning behavior and supervisor perceived status threat ($\chi^2 = 624.40$, $df = 154$, $p < .001$; CFI = 0.90, RMSEA = 0.10, SRMR_(within) = 0.17, SRMR_(between) = 0.17); (2) supervisor perceived status threat and abusive supervisory behavior ($\chi^2 = 607.62$, $df = 154$, $p < .001$; CFI = 0.91, RMSEA = 0.10, SRMR_(within) = 0.14, SRMR_(between) = 0.19); (3) all variables measured at Time 1 (i.e., subordinate boundary spanning behavior, supervisor managerial self-efficacy, and supervisor perceived status threat) ($\chi^2 = 952.29$, $df = 157$, $p < .001$; CFI = 0.83, RMSEA = 0.13, SRMR_(within) = 0.17, SRMR_(between) = 0.42). These results (see Table 2) demonstrated the discriminant validity of the constructs.

3.1.4.2. Hypotheses testing. Table 3 presents the results of the MSEM model examining all the hypotheses. Hypothesis 1 predicted that subordinate boundary spanning behavior and supervisor managerial self-efficacy would have an interaction effect on supervisor perceived status threat. The results showed that supervisor managerial self-efficacy was negatively and significantly related to the within-dyad random slope between subordinate boundary spanning behavior and supervisor perceived status threat ($\gamma = -0.16$, $p = .01$). Fig. 2 presents the interaction effect and simple slope analyses (Preacher et al., 2006), demonstrating that with low levels of supervisor managerial self-efficacy (-1 SD), subordinate boundary spanning behavior was positively and significantly related to supervisor perceived status threat (*simple slope* = 0.18, $p = .00$), but with high levels ($+1$ SD), subordinate boundary spanning behavior was not related to supervisor perceived status threat (*simple slope* = -0.07 , $p = .20$). Thus, Hypothesis 1 was supported in Study 1. Hypothesis 2 predicted that supervisor perceived status threat would be positively related to abusive supervisory behavior. The results showed that after we controlled for company dummy and industry dummy variables, the perceived status threat of supervisors was positively and significantly related to abusive supervisory behavior ($\gamma = 0.14$, $p = .04$), thus supporting Hypothesis 2.

Hypothesis 3 predicted that supervisor managerial self-efficacy

Table 1

Means, standard deviations, and correlations of the variables in Study 1.

Variables	M	SD	1	2	3	4	5
Subordinate level							
1. Subordinate boundary spanning behavior	4.10	1.25					
2. Supervisor perceived status threat	1.88	0.95	0.12*				
3. Abusive supervisory behavior	1.34	0.49	0.12*	0.32**			
Supervisor level							
1. Company dummy 1	0.71	0.46					
2. Company dummy 2	0.12	0.33	-0.58^{**}				
3. Industry dummy 1	0.23	0.42	0.14	0.02			
4. Industry dummy 2	0.25	0.44	0.12	0.07	-0.32^{**}		
5. Industry dummy 3	0.20	0.40	-0.01	-0.19	-0.28^{**}	-0.29^{**}	
6. Supervisor MSE	5.51	0.78	0.22*	-0.03	0.04	0.02	0.02

Note. $N = 320$ at the subordinate level; $N = 99$ at the supervisor level; MSE = managerial self-efficacy; * $p < .05$; ** $p < .01$

Table 2

Comparison of measurement models in Study 1.

Models	χ^2	df	$\Delta\chi^2$	CFI	RMSEA	SRMR _(within)	SRMR _(between)
Baseline model	166.83	149		1.00	0.02	0.03	0.07
Model 1: combine SBS and SPST;	624.40	154	462.26**	0.90	0.10	0.17	0.17
Model 2: combine SPST and AS;	607.62	154	439.99**	0.91	0.10	0.14	0.19
Model 3: combine all variables measured at Time 1 (i.e., SBS, SPST and SMSE)	952.29	157	784.20**	0.83	0.13	0.17	0.42

Note. SBS = Subordinate boundary spanning behavior; SPST = Supervisor perceived status threat; SMSE = Supervisor managerial self-efficacy; ** $p < .01$

Table 3

Unstandardized coefficients of MSEM for hypotheses testing in Study 1.

Variables	Supervisor perceived status threat		Abusive supervisory behavior	
	Est.	SE	Est.	SE
Subordinate boundary spanning behavior	0.06	0.05	0.01	0.03
Supervisor MSE	-0.17	0.11	0.14	0.08
Subordinate boundary spanning behavior ^(group-mean)	0.22*	0.10	0.03	0.05
Cross-level interactions				
Subordinate boundary spanning behavior × Supervisor MSE	-0.16*	0.07	-0.00	0.04
Supervisor perceived status threat × Supervisor MSE			-0.11**	0.04
Supervisor perceived status threat			0.14*	0.07
Company dummy	Included		Included	
Industry dummy	Included		Included	
Pseudo R^2	0.15		0.38	

Note. $N = 320$ at the subordinate level, $N = 99$ at the supervisor level; MSEM = multilevel structural equation modeling; SE = standard error; MSE = managerial self-efficacy; $p^* < .05$; $p^{**} < .01$

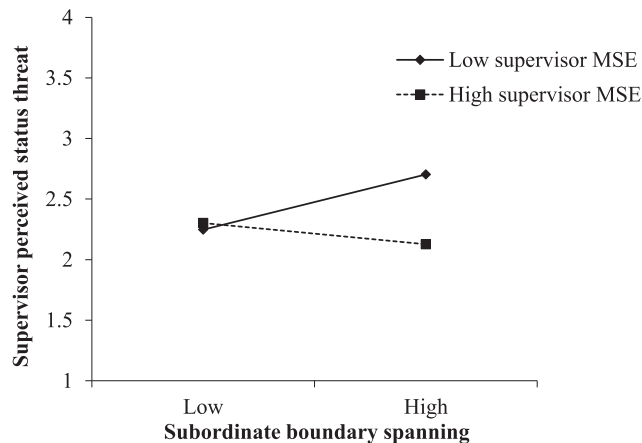


Fig. 2. Interaction effect of subordinate boundary spanning behavior and supervisor managerial self-efficacy on supervisor perceived status threat in Study 1.

attenuates the positive relation between supervisor perceived status threat and abusive supervisory behavior. The results showed that managerial self-efficacy was negatively and significantly related to the within-dyad random slope between supervisor perceived status threat and abusive supervisory behavior ($\gamma = -0.11, p = .01$). The simple slope analyses demonstrated that supervisor perceived status threat was positively and significantly related to abusive supervisory behavior when supervisor managerial self-efficacy was lower (-1 SD; simple slope = 0.22, $p = .00$), but not significant when managerial self-efficacy was higher ($+1$ SD; simple slope = 0.06, $p = .35$). Thus Hypothesis 3 was supported in Study 1.

Hypothesis 4 predicted that supervisor perceived status threat would

mediate the negative interactive effect of subordinate boundary spanning behavior and supervisor managerial self-efficacy on abusive supervisory behavior. We further tested the conditional indirect effect for abusive supervisory behavior. The results showed that at a lower level of supervisor managerial self-efficacy (-1 SD), the indirect effect was significant (95% CI [0.001, 0.05]), whereas, at a higher level ($+1$ SD), the indirect effect was not significant (95% CI [-0.04, 0.01]). The difference between these two conditional indirect effects was also significant (95% CI [-0.06, -0.01]). Thus, Hypothesis 4 was supported in Study 1. Predictors included in our model accounted for 15% of the total variance in supervisor perceived status threat and 38% of the total variance in abusive supervisory behavior, respectively.

It is worth noting that the distributions of supervisor perceived status threat and abusive supervisory behavior were positively skewed due to their low base-rate feature (e.g., Khan et al., 2018; Mitchell & Ambrose, 2007; Tepper, 2000). In supplementary analyses, we used square-root-transformed dependent variables to correct for their skewness (e.g., Song et al., 2018). The results of the analyses were virtually consistent with the results reported in this manuscript.

All the hypothesized relationships received support in Study 1. However, the design had limitations. Study 1 ignored certain factors that may affect supervisor perceived status threat as well as abusive supervisory behavior, such as subordinates' proactive personality (e.g., Parker et al., 2019), job performance (e.g., Khan et al., 2018; Xu et al., 2012), and supervisors' resource depletion (e.g., Barnes et al., 2015). Controlling for these confounding factors and alternative explanations provides more compelling support for our hypotheses. Therefore, we conducted another time-lagged, multiple-source field study (i.e., Study 2).

3.2. Study 2: Methods

3.2.1. Participants and procedure

We collected data at two companies operating in silicon chip manufacturing and financial software industries, respectively, in eastern China. The HR managers provided us with lists of team leaders and members. Although the specific functions of these teams varied, all of them were responsible for providing particular products or services, and they engaged to different degrees in external activities both within and outside their organizations.

We collected data in two waves at an interval of 2 weeks to reduce potential biases induced by common methods (Podsakoff et al., 2012). Our sample initially consisted of 108 supervisors and 403 immediate employees. With the help of HR managers, at Time 1, we separately distributed online survey links to the subordinates and supervisors. In the first wave, 347 subordinates completed the measure of boundary spanning behavior, resulting in a response rate of 86.1%. In the same wave, 93 supervisors completed measures of managerial self-efficacy and perceived status threat from each participating subordinate, resulting in a response rate of 86.1%. At Time 2, out of the 403 immediate employees that we invited to participate in the study, 306 subordinates returned the survey on abusive supervisory behavior, yielding a response rate of 79.2%. We used the full-information maximum likelihood (FIML) approach (Newman, 2014), which is Mplus's default approach to datasets with missing data (Muthén & Muthén, 1998–2017). After matching the responses, we finally obtained usable

data from 93 supervisors and 328 immediate subordinates.

Of the 93 supervisors, 86.0% were male, and 80.7% had a college or higher education; they had an average age of 36.3 years ($SD = 5.4$) and an average organizational tenure of 9.4 years ($SD = 4.3$). Of the 328 subordinates, 79.3% were male, and 73.5% had a college or higher education; they had an average age of 31.7 years ($SD = 5.6$) and an average organizational tenure of 5.4 years ($SD = 3.4$). The average relationship tenure between the supervisors and subordinates was 2.9 years ($SD = 2.3$).

3.2.2. Measures

As in Study 1, the measures used in Study 2 were translated from English into Chinese following Brislin's (1980) translation-back translation procedure. For subordinate boundary spanning behavior ($\alpha = 0.93$), supervisor managerial self-efficacy ($\alpha = 0.90$), supervisor perceived status threat ($\alpha = 0.92$), and abusive supervisory behavior ($\alpha = 0.94$), we used the same scales as in Study 1.

Control variables. Drawing on the literature, we controlled for several variables. As our data were collected from two different companies, we controlled for the potential systematic effects of company type by creating a dummy variable. In addition, we included three more control variables, and all of them were scored on a 7-point Likert scale (1 = *strongly disagree* and 7 = *strongly agree*).

First, individuals' proactive personality might be related to their discretionary boundary spanning behavior (e.g., Parker et al., 2019). To reduce the confounding influence of this personality trait on supervisor perceived status threat and abusive supervisory behavior, we included subordinates' proactive personality as a control variable and measured it with a 4-item scale adopted from Wu et al. (2018). Second, while some research has suggested that subordinates with poor performance are more likely to be abused (e.g., Mackey et al., 2017; Xu et al., 2012), other research has shown that high performers may also engender abusive behavior due to supervisors' perceptions of threats (Khan et al., 2018). Therefore, we included subordinates' performance as a control variable and asked supervisors to report each subordinate's performance. To reduce supervisors' burden and prevent response fatigue (e.g., Lam et al., 2011; Tse et al., 2013), we chose the two items with the highest factor loadings from Gong et al. (2009) to capture subordinate performance: "This person always completes job assignments on time" and "This person's work performance always meets the expectations of the supervisor" (the factor loadings were 0.99 and 0.95, respectively; see Gong et al., 2009, p. 769). Third, research has suggested that supervisors who lack self-regulatory resources are more likely than others to be abusive (e.g., Barnes et al., 2015; Tepper et al., 2017), so we controlled for supervisors' resource depletion with a 5-item scale adopted from Johnson et al. (2014). As in Study 1, our analyses revealed that none of the subordinates' or supervisors' demographic control variables significantly altered our results, and thus we did not retain any demographic controls in our final analysis (Becker, 2005; Carlson & Wu, 2012). The analyses with and without control variables produced virtually identical results.

3.2.3. Analytical strategy

As in Study 1, we conducted a two-level path analysis within the random slope framework of MSEM (Liu et al., 2012; Preacher et al., 2016) using Mplus 8.3 (Muthén & Muthén, 1998–2017). Following Hofmann and Gavin (1998), we included the group-mean of subordinate boundary spanning behavior at the between level in the estimations of the random intercept of the endogenous variable. We adopted the same procedure used in Study 1 to investigate our four hypotheses.

3.2.4. Study 2: Results

3.2.4.1. Descriptive statistics and confirmatory factor analyses. We first estimated null models with only supervisor perceived status threat or

abusive supervisory behavior as the dependent variable. The results showed significant between-team variance in supervisor perceived status threat ($\chi^2 = 427$, $df = 92$, $p < .001$; ICC1 = 0.50, indicating that 50% of variance resided between teams). While the between-team variance of abusive supervisory behavior is not significant ($\chi^2 = 107$, $df = 88$, $p = .08$; ICC1 = 0.06, indicating that 6% of variance resided between teams). The descriptive statistics and correlations of the study variables are presented in Table 4.

As in Study 1, we examined the distinctiveness of our main constructs by conducting the multilevel CFA using the procedures recommended by Dyer et al. (2005). To achieve a more optimal variable-to-sample-size ratio and minimize parameter estimate instability, we used the item-to-construct-balance parceling method which is similar to Study 1 (Bagozzi & Edwards, 1998; Little et al., 2002). We followed Study 1 and assigned subordinate boundary spanning and supervisor managerial self-efficacy items into four parcels, respectively and abusive supervisory behavior items into five parcels by sequentially averaging the aggregated items with the highest and lowest loadings (Little et al., 2002; Little et al., 2013). We used Mplus 8.3 (Muthén & Muthén, 1998–2017) to test the model in which supervisor managerial self-efficacy was treated as a group-level factor, and the other three constructs were treated as both within-dyad and group-level factors. Results showed that the hypothetical four-factor baseline model fit the data well ($\chi^2 = 173.67$, $df = 149$, $p = .08$; CFI = 0.99, RMSEA = 0.02, SRMR_(within) = 0.03, SRMR_(between) = 0.12). This model fitted the data better than alternative models when the following variables were combined: (1) subordinate boundary spanning behavior and supervisor perceived status threat ($\chi^2 = 633.10$, $df = 154$, $p < .001$; CFI = 0.87, RMSEA = 0.10, SRMR_(within) = 0.16, SRMR_(between) = 0.22); (2) supervisor perceived status threat and abusive supervisory behavior ($\chi^2 = 613.25$, $df = 154$, $p < .001$; CFI = 0.87, RMSEA = 0.10, SRMR_(within) = 0.14, SRMR_(between) = 0.29); (3) all constructs measured at Time 1 (i.e., subordinate boundary spanning behavior, supervisor managerial self-efficacy, and supervisor perceived status threat) ($\chi^2 = 918.46$, $df = 157$, $p < .001$; CFI = 0.79, RMSEA = 0.12, SRMR_(within) = 0.16, SRMR_(between) = 0.27). These results (see Table 5) demonstrated the discriminant validity of the constructs.

3.2.4.2. Hypotheses testing. Table 6 presents the results of the MSEM model examining all the hypotheses. Hypothesis 1 predicted that subordinate boundary spanning behavior and supervisor managerial self-efficacy would have an interaction effect on supervisor perceived status threat. The results showed that the moderating effect of supervisor managerial self-efficacy was negative and significant ($\gamma = -0.31$, $p = .02$). Fig. 3 presents the interaction effect and simple slope analyses (Preacher et al., 2006), demonstrating that with low levels of supervisor managerial self-efficacy ($-1 SD$), subordinate boundary spanning was positively and significantly related to supervisor perceived status threat (simple slope = 0.30, $p < .001$), and with high levels ($+1 SD$), subordinate boundary spanning behavior was negatively and significantly related to supervisor perceived status threat³ (simple slope = -0.12 , $p = .03$). Thus, Hypothesis 1 was supported in Study 2. Hypothesis 2 predicted that supervisor perceived status threat would be positively related to abusive supervisory behavior. The results showed that after controlling for company type, subordinate proactive personality, subordinate performance, and supervisor resource depletion, the perceived status threat of supervisors was positively and significantly related to abusive supervisory behavior ($\gamma = 0.22$, $p < .001$), thus supporting Hypothesis 2.

³ We should be cautious when interpreting the negative relationship between subordinate boundary spanning behavior and supervisor perceived status threat at a high level of supervisor managerial self-efficacy because (a) this finding was not consistent across different studies and (b) the relationship became nonsignificant when the control variables were excluded from the model (simple slope = -0.08 , $p = .07$).

Table 4

Means, standard deviations, and correlations of the variables in Study 2.

Variables	M	SD	1	2	3	4	5
Subordinate level							
1. Subordinate proactive personality	4.76	1.01					
2. Subordinate performance	4.72	1.21	0.06				
3. Company dummy	0.37	0.48	0.19**	0.03			
4. Subordinate boundary spanning behavior	4.49	0.95	0.39**	−0.05	−0.19**		
5. Supervisor perceived status threat	1.86	0.92	−0.00	−0.03	0.09	0.02	
6. Abusive supervisory behavior	1.37	0.47	−0.03	−0.07	0.03	−0.04	0.16**
Supervisor level							
1. Supervisor MSE	5.79	0.67					
2. Supervisor resource depletion	2.51	0.76	−0.33**				

Note. $N = 328$ at the subordinate level; $N = 93$ at the supervisor level; MSE = managerial self-efficacy; ** $p < .01$ **Table 5**

Comparison of measurement models in Study 2.

Models	χ^2	df	$\Delta\chi^2$	CFI	RMSEA	SRMR _(within)	SRMR _(between)
Baseline model	173.67	149		0.99	0.02	0.03	0.12
Model 1: combine SBS and SPST	633.10	154	459.65**	0.87	0.10	0.16	0.22
Model 2: combine SPST and AS	613.25	154	444.99**	0.87	0.10	0.14	0.29
Model 3: combine SBS, SPST and SMSE	918.46	157	743.09**	0.79	0.12	0.16	0.27

Note. SBS = Subordinate boundary spanning behavior; SPST = Supervisor perceived status threat; SMSE = Supervisor managerial self-efficacy; ** $p < .01$ **Table 6**

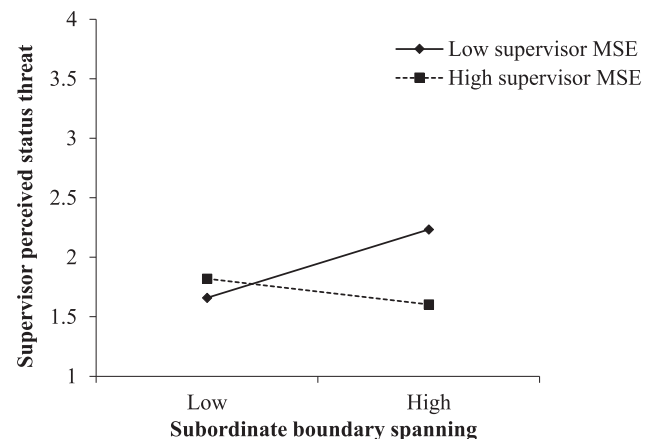
Unstandardized coefficients of MSEM for hypotheses testing in Study 2.

Variables	Supervisor perceived status threat		Abusive supervisory behavior	
	Est.	SE	Est.	SE
Company dummy	0.11	0.16	0.01	0.06
Subordinate proactive personality	0.04	0.05	−0.02	0.03
Subordinate performance	−0.03	0.04	−0.01	0.03
Supervisor resource depletion	0.16	0.10	−0.03	0.04
Subordinate boundary spanning behavior	0.09	0.07	0.00	0.05
Supervisor MSE	−0.18	0.11	0.02	0.12
Subordinate boundary spanning behavior ^(group-mean)	−0.09	0.12	−0.02	0.07
Cross-level interactions				
Subordinate boundary spanning behavior × Supervisor MSE	−0.31*	0.14	−0.09	0.07
Supervisor perceived status threat × Supervisor MSE			−0.02	0.08
Supervisor perceived status threat			0.22***	0.06
Pseudo R^2	0.10		0.11	

Note. $N = 328$ at the subordinate level, $N = 93$ at the supervisor level; MSEM = multilevel structural equation modeling; SE = standard error; MSE = managerial self-efficacy; $p^* < .05$; $p^{***} < .001$.

Hypothesis 3 predicted that supervisor managerial self-efficacy attenuates the positive relation between supervisor perceived status threat and abusive supervisory behavior. The results showed that managerial self-efficacy was not related to the within-dyad random slope between supervisor perceived status threat and abusive supervisory behavior ($\gamma = -0.02$, $p = .81$). Thus, **Hypothesis 3** was not supported in Study 2.

Hypothesis 4 predicted that supervisor perceived status threat would mediate the negative interactive effect of subordinate boundary spanning behavior and supervisor managerial self-efficacy on abusive supervisory behavior. We further tested the conditional indirect effect for abusive supervisory behavior. The results showed that at a lower level of supervisor managerial self-efficacy (-1 SD), the indirect effect was positive and significant (95% CI [0.01, 0.11]), and at a higher level of supervisor managerial self-efficacy ($+1$ SD), the indirect effect was not significant (95% CI [−0.09, 0.02]). The difference between these two conditional indirect effects was significant (95% CI [−0.12, −0.06]). Thus, **Hypothesis 4** was supported in Study 2. Predictors included in our model

**Fig. 3.** Interaction effect of subordinate boundary spanning behavior and supervisor managerial self-efficacy on supervisor perceived status threat in Study 2.

accounted for 10% of the total variance in supervisor perceived status threat and 11% of the total variance in abusive supervisory behavior, respectively. In supplementary analyses, we used square-root-transformed dependent variables to correct for the skewness of dependent variables (e.g., Song et al., 2018). The results of the analyses are virtually consistent with the results reported in this manuscript.

Studies 1 and 2 consistently supported most of the hypotheses using field data. Although Studies 1 and 2 were high in external validity, they were not suitable for making causal inferences. Therefore, a controlled experiment was conducted to address this constraint. According to Aguinis and Bradly (2014), vignette-style experiments consist of carefully constructed and realistic scenarios that aid in establishing the causality of research findings while keeping a relatively high level of realism. Furthermore, vignette designs are appropriate and popular for experimental studies of abusive supervision, because they avoid violations of research ethics (e.g., Fischer et al., 2021). Therefore, we conducted an experimental study (i.e., Study 3) using vignette designs.

3.3. Study 3: Methods

3.3.1. Participants and design

We recruited 184 adults working full-time to participate in our research in exchange for US\$1.30 through Sojump, a Chinese online scientific research platform created to facilitate social science studies with human subjects (Chua et al., 2019). We used a 2 (supervisor managerial self-efficacy: high vs. low) $\times$ 2 (subordinate boundary spanning: high vs. low) between-person design and randomly assigned participants to one of the four conditions. We excluded eight participants who failed the attention checks. As our research examined supervisory responses to subordinate behavior, we followed the approach used in other studies (e.g., Gauglitz et al., 2023; Schmitt et al., 2022) and retained participants who occupied managerial positions.⁴ All of our analyses were conducted using the final sample of 150 participants (52.7% female; average age = 31.0 years, $SD = 5.8$; average work tenure = 7.8 years, $SD = 5.1$), 92.7% of whom had a bachelor's degree. The participants were from the manufacturing, services, trade, financial, information technology, real estate, and other industries.

3.3.2. Experimental procedure and materials

The participants were instructed to consider themselves the manager of a product team in a manufacturing company. The participants read the following scenario:

Company A is a manufacturing company that has set up a committee to supervise the establishment and development of new products. You are the leader of a team that is mainly engaged with the development of a new product. Once a new product development project is set up, your team needs to connect and coordinate with other function teams (for example, the testing team, marketing team, etc.) to jointly complete the project. Your team also needs to frequently report the project's progress to the committee to obtain support for further product development. During this time, you are tasked with managing the project's progress, coordinating activities, and acquiring the necessary resources.

We then provided background information about the manager's past work experience in this team to manipulate the participants' perceptions of managerial self-efficacy, and then presented a vignette of one team member's work behavior to manipulate the participants' perceptions of the subordinate boundary spanning behavior. After providing the manipulation information, we asked the participants to respond to the perceived status threat measure and complete the measure of abusive supervisory behavior.

Managerial self-efficacy manipulation. Following Fast et al. (2014), we manipulated managerial self-efficacy by describing the manager's performance in the specific position in the scenario. The participants in the high managerial self-efficacy condition read the following:

To date, things have been working out just as you had planned: over the past 2 years, profits have increased and team morale has been strengthened. Top executives have recognized that you are a highly competent product team leader.

The participants in the low managerial self-efficacy condition read the following:

To date, things are not working out as you had planned: over the past 2 years, profits have declined and team morale has deteriorated. Top executives have begun to question whether you have what it takes to be a competent product team leader.

To check the effectiveness of the manipulation, we asked the participants to report their perceived levels of managerial self-efficacy using

an 8-item scale adapted from Fast et al. (2014). A sample item is "In this team, I believe I can succeed at almost any endeavor to which I set my mind." They rated their agreement with each statement on a 7-point response scale (1 = *strongly disagree* and 7 = *strongly agree*, $\alpha = 0.92$).

Subordinate boundary spanning behavior manipulation. To manipulate subordinate boundary spanning behavior, we first explained to the participants that one of their team members, Le Li (a gender-neutral name in the Chinese context), worked on product design and development tasks. We then presented manipulation information developed from previous investigations into boundary spanning behavior (Ancona & Caldwell, 1992; Faraj & Yan, 2009). The participants in the boundary spanning behavior condition were informed that.

Recently, you have noticed that Le Li often confers with colleagues from other teams about product development on behalf of the focal team and seeks resources and support from the product committee.

The participants in the control condition were informed that.

Recently, you have noticed that Le Li often informs other teammates of the problem he/she has encountered and seeks help and support from other team members before resolving task problems alone.

To check the effectiveness of the manipulation, we asked participants to report subordinate Le Li's boundary spanning behavior using a 4-item scale adapted from Faraj and Yan (2009). This scale was originally used to measure team boundary spanning; we modified the items and made them more applicable to the individual boundary spanning behavior in our scenario. A sample item is "Le Li often solicits information and resources from outside the team." The participants rated their agreement with each statement via a 7-point response scale (1 = *strongly disagree* and 7 = *strongly agree*, $\alpha = 0.78$).

3.3.3. Measures

Supervisor perceived status threat. As in Study 1, we measured supervisors' perceptions of status threat with the 3-item scale adapted from Burris (2012, $\alpha = 0.88$). A sample item is "My ability to lead the team will be questioned because of Le Li's behavior."

Abusive supervisory behavior. We used the 5-item scale from Mitchell and Ambrose (2007) for brevity. This shortened scale captures the active interpersonal abusive behavior of the supervisor (Mitchell & Ambrose, 2007) and is the most consistent with our argument that abusive supervisory behavior can damage the status of the target subordinate. To avoid breaching research ethics and to eliminate social desirability concerns, we also adopted an indirect measure to capture participants' tendency to enact abusive supervisory behavior (e.g., Lam et al., 2017; Walter et al., 2015). The participants reported the extent to which they agreed it was acceptable to treat Le Li in various ways, using a 7-point response scale (1 = *strongly disagree* and 7 = *strongly agree*, $\alpha = 0.89$). Sample items are "Ridicules Le Li" and "Puts Le Li down in front of others."

3.3.4. Study 3: Results

3.3.4.1. Manipulation checks. We conducted a *t*-test, which showed that supervisor managerial self-efficacy was rated significantly higher in the high ($M = 5.74$, $SD = 0.63$) rather than the low supervisor managerial self-efficacy condition ($M = 4.65$, $SD = 1.14$), $t(148) = -7.25$, $p < .001$. Subordinate boundary spanning behavior was rated significantly higher in the boundary spanning behavior condition ($M = 5.36$, $SD = 0.64$) than in the control condition ($M = 4.47$, $SD = 1.15$), $t(148) = -5.82$, $p < .001$. These results indicated that our manipulations were successful.

3.3.4.2. Hypotheses testing. The descriptive statistics and correlations of the study variables are presented in Table 7. We used Model 59 of Hayes's (2022) PROCESS macro (version 4.3) with SPSS 23.0 to test the hypotheses of Study 3. Table 8 presents the results of the path analysis

⁴ We conducted the supplementary analysis with 176 participants, including participants who did not hold managerial positions. The results were virtually identical to those reported in this manuscript.

Table 7

Means, standard deviations, and correlations of the variables in Study 3.

Variables	M	SD	1	2	3
1. Subordinate boundary spanning behavior	0.48	0.50			
2. Supervisor MSE	0.51	0.50	0.01		
3. Supervisor perceived status threat	3.40	1.32	0.36**	-0.23**	
4. Abusive supervisory behavior	2.44	1.16	0.05	-0.11	0.50**

Note. $N = 150$. MSE = managerial self-efficacy; ** $p < .01$ **Table 8**

Unstandardized coefficients of path analysis for hypotheses testing in Study 3.

Variables	Supervisor perceived status threat		Abusive supervisory behavior	
	Est.	SE	Est.	SE
Subordinate boundary spanning behavior	1.44***	0.28	-0.49	0.27
Supervisor MSE	-0.16	0.27	0.55	0.47
Subordinate boundary spanning behavior $\times$ Supervisor MSE	-0.93*	0.39	0.19	0.36
Supervisor perceived status threat $\times$ Supervisor MSE			-0.18	0.14
Perceived status threat			0.57***	0.09
R^2	0.22		0.28	

Note. $N = 150$. MSE = managerial self-efficacy; * $p < .05$; *** $p < .001$

examining all the hypotheses. **Hypothesis 1** predicted an interaction effect of subordinate boundary spanning behavior and supervisor managerial self-efficacy on supervisor perceived status threat. The results showed that the interaction term was negatively and significantly related to perceived status threat ($b = -0.93, p = .02$). Fig. 4 illustrates the results of a simple slope test. When supervisor managerial self-efficacy was low, the boundary spanning behavior was positively and significantly related to the perceived status threat ($b = 1.44, p < .001$), whereas when it was high, no relationship was found ($b = 0.51, p = .06$). Therefore, **Hypothesis 1** was supported in Study 3. **Hypothesis 2** predicted that the perceived status threat of supervisors would be positively related to their abusive supervisory behavior. The results showed that perceived status threat was positively related to abusive supervisory behavior ($b = 0.57, p < .001$), and thus **Hypothesis 2** was supported in Study 3.

Hypothesis 3 predicted that supervisor managerial self-efficacy attenuates the positive association between supervisor perceived status threat and abusive supervisory behavior. The results showed that the interaction term was not related to abusive supervisory behavior ($b =$

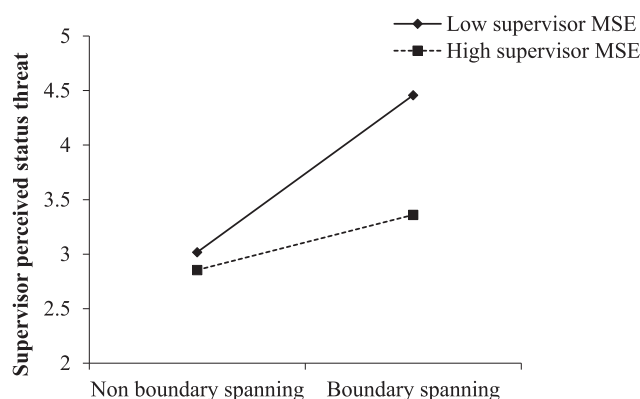


Fig. 4. Interaction effect of subordinate boundary spanning behavior and supervisor managerial self-efficacy on supervisor perceived status threat in Study 3.

-0.18, $p = .22$). Thus, **Hypothesis 3** was not supported in Study 3.

Hypothesis 4 predicted that supervisor perceived status threat would mediate the negative interactive effect of subordinate boundary spanning behavior and supervisor managerial self-efficacy on abusive supervisory behavior. Results showed that when supervisor managerial self-efficacy (-1 SD) was low, the indirect effect was significant (indirect effect = 0.83, 95% CI [0.43, 1.35]), while at higher level of supervisor managerial self-efficacy (+1 SD) it was non-significant (indirect effect = 0.20, 95% CI [-0.003, 0.45]). The differences in the degrees of these effects were also significant (conditional indirect effect = -0.62; 95% CI [-1.19, -0.15]). Thus, **Hypothesis 4** was supported in Study 3.

4. General discussion

Based on self-protection theory, we proposed and examined a boundary condition and psychological mechanism underlying supervisors' negative responses to their subordinates' boundary spanning behavior. In our three studies (two survey studies and one experimental study), we found that perceived status threats and abusive supervisory behavior, as responses to subordinates' boundary spanning, are contingent on the supervisors' managerial self-efficacy. We discuss the theoretical and practical implications of our findings below.

4.1. Theoretical contributions

Our research makes several contributions to the literature. First, we contribute to the boundary spanning literature by revealing which supervisors are likely to respond negatively to subordinates' boundary spanning behavior. Recent research has revealed that subordinate boundary spanning often results in supervisors' threatening perceptions and negative supervisory responses (e.g., Clement et al., 2018; Mell et al., 2022). For instance, Mell et al. (2022) indicated that supervisors are prone to respond to frequent boundary spanning by subordinates by perceiving a loss of territorial control and exhibiting undermining behaviors. They also found that subordinates who cross the team boundary while seeking advice from their supervisors are less susceptible to supervisory undermining (Mell et al., 2022). However, these studies have largely overlooked the fact that not all supervisors respond negatively to subordinates who engage in boundary spanning. The failure to consider the role of supervisors' characteristics has created an incomplete picture of when and why supervisors become internal barriers to subordinates seeking to span team boundaries. Our research illustrates that supervisors who are not confident of their ability to carry out their managerial role expectations or those with lower managerial self-efficacy are more likely to perceive subordinates' behavior as threatening and to have adverse reactions to subordinates' boundary spanning.

Second, we advance the literature by uncovering a critical underlying mechanism driving the relationship between subordinates' boundary spanning and negative supervisory responses. Other studies of supervisors' adverse reactions have relied on territoriality theory and viewed team members' boundary spanning as transgressions that undermine supervisors' control over the group territory (Mell et al., 2022). However, these studies have failed to recognize that boundary spanning is a critical function for team effectiveness, as well as an essential expectation associated with supervisors' managerial roles (e.g., Morgeson et al., 2010). The extent to which supervisors carry out these role expectations has been significantly associated with their social status at work (Fast et al., 2014; Hollander, 1958). In light of this insight, we drew on self-protection theory (Sedikides, 2012; Sedikides & Alicke, 2012) and identified status as a significant psychological driver that triggers abusive supervisory behavior in response to subordinates' frequent boundary spanning, particularly when they have a low sense of managerial self-efficacy.

Third, we contribute to self-protection theory (Sedikides, 2012; Sedikides & Alicke, 2012) by using this fundamental framework to explore power holders' self-protection. Our findings reveal how

subordinates' behaviors, even those that benefit the team, might inadvertently pose a potential threat to the power holders' social status. Thus, we identify a threat to status as an essential type of threat that supervisors may perceive. We also recognize abusive supervisory behavior as an aggressive self-protection strategy. Cognitive tuning and disengagement have been identified as effective strategies for coping with self-threats (e.g., [Chen et al., 2013](#); [Green et al., 2008](#); [Mitchell et al., 2018](#)). For example, individuals may selectively forget feedback that threatens their self-regard ([Green et al., 2008](#)) or withdraw from a task to protect their sense of self from further mistreatment ([Chen et al., 2013](#)). In this research, we reveal that aggressive interpersonal behavior (i.e., abusive supervisory behavior) also can be a self-protection strategy, particularly for power holders.

Fourth, we extend the literature on abusive supervision. [Tepper et al. \(2017\)](#) have concluded that "abusive supervision may be a reaction to provocative subordinate behavior" (p. 137). Yet, the provocative behaviors examined in previous research were mainly negative behaviors that interfere with supervisors' goal accomplishment or authority, such as poor performance (e.g., [Liang et al., 2016](#); [Tepper et al., 2011](#); [Wang et al., 2015](#)) and deviance directed at the organization (e.g., [Lian et al., 2014](#)) or supervisor (e.g., [Simon et al., 2015](#)). An exception is a study by [Khan et al. \(2018\)](#), which showed that high performance can induce abusive supervision in some circumstances. Our research demonstrates that subordinates who engage in the team function of boundary spanning could also become the targets of abusive supervision, because supervisors with low managerial self-efficacy may perceive subordinates' boundary spanning as a threat to their status.

4.2. Practical implications

Our findings suggest that there are practical strategies organizations can use to ensure that they benefit from team members' boundary spanning behavior. First, team supervisors should be aware that subordinates' external networks and voluntary actions are valuable team assets ([Ancona & Bresman, 2007](#)). Abusive supervisory behavior causes subordinates who span the team boundary to feel devalued and less willing to span boundaries in the future, which can be detrimental to teams, particularly those with supervisors who have low managerial self-efficacy as their confidence in fulfilling their managerial role may further diminish. Thus, we suggest that supervisors with low managerial self-efficacy should be encouraged to recognize their inherent self-protection motives and the source of their negative perceptions of subordinates' boundary spanning behavior, which will discourage them from mistreating these boundary spanning subordinates.

Second, upper-level managers should help supervisors develop their managerial self-efficacy. Our research shows that when supervisors are confident of their abilities to fulfill their managerial responsibilities, they are less likely to be threatened by subordinates' boundary spanning behavior and hence less likely to engage in "bad" leadership behavior. Training opportunities such as leadership development workshops can be conducted to improve their leadership skills, and more timely positive feedback can be provided to supervisors, thus increasing their self-assurance.

Finally, team members should recognize that supervisors may feel threatened by their boundary spanning behavior, particularly when supervisors lack managerial self-efficacy. To avoid mistreatment from supervisors, team members should consider whether their supervisors are confident of their managerial capabilities before engaging in boundary spanning behavior. The team members can then present their boundary spanning activities appropriately to make them more acceptable in the eyes of their supervisors. For example, a team member can inform their supervisor about information received from external players and allow the supervisor to decide how to introduce the input to the team.

4.3. Limitations and future directions

Despite our insightful findings, this research has several limitations that may indicate new directions for future research. First, the attenuating effect of managerial self-efficacy on the positive association between supervisor perceived status threat and abusive supervision was inconsistent across the three studies. This may stem from the fact that status represents one of the highest levels of personal interests ([Alicke & Sedikides, 2009](#); [Sedikides, 2012](#)), particularly for supervisors. Supervisors are, therefore, inclined to respond promptly or even aggressively to perceived status threats. Nevertheless, research has also indicated that the general concern for status might vary between individuals ([Blader & Chen, 2011](#)). Future research could explore how other critical characteristics, such as supervisors' concern for status, might alter their attribution of subordinates' boundary spanning, as well as their behavioral responses.

Second, we did not consider the environmental characteristics that may function as important contingencies. A turbulent environment and complicated cross-function tasks may require teams to consider their external processes (e.g., [Choi, 2002](#); [Marrone, 2010](#); [Richter et al., 2006](#); [Yan & Louis, 1999](#)). For instance, [Chen et al. \(2018\)](#) found that team members' perceptions of technological turbulence prompts them to depend less on experts and to strive to acquire new knowledge outside the team. In such turbulent environments, supervisors may be more likely to recognize the necessity and value of subordinates' boundary spanning behavior. Thus, we suggest that future research should examine how external contextual factors affect supervisors' reactions to subordinates' boundary spanning behavior.

Third, we only explored the conditions under which supervisors are likely to respond negatively to their subordinates' boundary spanning behavior. In two field studies (i.e., Studies 1 and 2), under a high level of managerial self-efficacy, there was a marginal and negative association between subordinates' boundary spanning and supervisor perceived status threat. These findings may imply that supervisors with higher managerial self-efficacy are more inclined to regard subordinates' boundary spanning positively. Indeed, as boundary spanning behavior can benefit a team ([Marrone, 2010](#)), supervisors may recognize its value and respond positively to subordinates who engage in such behavior under certain circumstances. Future research could draw on new theoretical perspectives and explore the conditions under which subordinates' boundary spanning behavior can induce positive supervisory reactions.

5. Conclusion

Drawing on self-protection theory, we explored when and why supervisors demonstrate abusive supervisory behavior toward subordinates who engage in boundary spanning behavior. We found that supervisors with low (rather than high) managerial self-efficacy perceive subordinates' boundary spanning behavior as a threat to their status and thus respond to those spanning boundaries with abusive supervisory behavior. Our results offer insights into which supervisors are more prone to have adverse psychological and behavioral responses to such behavior. We encourage future research to extend our understanding of this phenomenon in increasingly dynamic and complex business environments.

CRediT authorship contribution statement

Xiao-Yun Xie: Writing – original draft, Validation, Supervision, Resources, Funding acquisition, Conceptualization. **Junjie Wei:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Qiongjing Hu:** Supervision, Funding acquisition, Conceptualization, Project administration, Writing – original draft, Writing – review & editing. **Zhenyu Liao:** Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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